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MAHATMA GANDHI UNIVERSITY , KOTTAYAM
MGU-UGP BCA (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
CORE COURSE (CC)- MG2CCRBCA100-MATHEMATICS FOUNDATION TO COMPUTER SCIENCE
(2024 ADMISSION ONWARDS)

Duration:2 Hours **Maximum Marks : 70**

SCHEME OF VALUATION

Remember (K), Understand (U) , Apply (A), Analyze (An), Evaluate E, Create (C), Skills(S), Interest (I) and Approximation(An)

Part A

Short answer type Questions

Answer any 5 questions. Each question carries 2 marks

1. Any two characteristics of the spanning tree- give 2 marks. (1 mark for each) [K]/[CO1]
2. Yes (1 mark). Reason- If all the vertices of a graph have even degree, then the graph has an Euler Circuit. (1 mark) [A]/[CO1]
3. 2 marks may be given for any values of h. [A]/[CO2]
4. Any 2 methods for finding the solution of a non-liner equation-(2 marks, 1 mark for each). [K]/[CO2]
   (Direct analytical methods, Graphical methods, trial and error methods and Iterative methods. Also give marks for different methods of each one . e.g. Newton -Raphson method, Bisection methods etc.)
5. Any 2 application –(2 marks, 1 mark for each). [U]/[CO3]
   (Defence operation, Industry, Planning, Agriculture, Transport)
6. Big M method is a modified simplex method for solving a LPP when a high penality cost or profit M has been assigned to the artificial variable in the objective function (or similar statement.(2 marks) [K]/[CO3]
7. Lowest cost entry method means choose the cell having lowest cost in the matrix. (similar statement) (2 marks). [K]/[CO4]
8. Steps for finding the initial feasible solution using Vogel's method or Explanation-(2 marks) [U]/[CO4]

(2X 5 =10)

Part B

Short Essay Type Questions.

Answer any 5 questions. Each question carries 6 marks

9. Definition of degree of a vertex - (2 marks). Degree of a self loop = 2 (1 mark). deg(A) =2, deg (B) = 4, deg(C) = 4, deg (D) = 4, deg (E) =4, deg (F) = 2. (½ mark for each one) (total 6 marks) [A]/[CO1]
10. a) Definition of tree –(2 marks), Ans. No (1 mark)
    b) Figure of trees with 3 vertices – (3 marks) [U]/[CO1]

11. For finding a=0, b=1 (1mark) $x_0 = \frac{a+b}{2} = \frac{1}{2} = 0.5$ (1 mark) .f(.5) = -0.625 (1 mark)

First approximate value $x_1 = \frac{.5+1}{2} = 0.75$ (1 mark) . f(0.75) = -0.0156 (1 mark).

Second approximate value $x_2 = \frac{0.75+1}{2} = 0.875$. (1 mark) .(Total 6 marks)

[A]/[CO2]

12.

Marks (x)	No, of students (y)	Δy	$\Delta^2 y$	
20	45			
30	115	70		
40	210	95	25	(table – 2 marks)

$u = \frac{x - x0}{h} = \frac{25-20}{10} = 0.5$ (1 mark)

$Y = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{1.2}\Delta^2 y + \ldots\ldots\ldots\ldots\ldots\ldots$ (1 mark)

$Y(25) = 76.875 \approx 77$ (substitution and answer – give 2 marks)

[K]/[CO2]

13. For finding co ordinates of equation – (1 mark) . Graphs of constraints (1 mark). Marking Feasible region (1 mark). Finding points (1 mark) substitution of points in Minimize $Z = 4x_1 + 5x_2$ (1 mark). Final answer = minimum z = 12 at (3,0) – (1 mark). Total 6 marks [A]/[CO3]
14. Formulation of equation :Minimize Z = 40x + 50y subject to $2x + y \leq 100$, $x + 3y \leq 150$,where as $x, y \geq 0$ give 2 marks . For finding co ordinates of equation – give 1 mark . Graphs of constraints (1 mark). Marking Feasible region (1 mark). Finding points and final answer – Maxi.Z = 3200 at (30,40) (1 mark). Total 6 marks [A]/[CO3]
15. Verification of balanced or not – (1 mark). Convert problem into balanced (1 mark) For finding initial feasible solution of any method. (4 marks). [U]/[CO4]
16. For finding initial feasible solution of any method. (4 marks) . steps of optimality solution (1 mark) ,final answer (1mark). [A]/[CO4]

(6X 5 = 30)

Part C

Essay Type Questions.

Answer any 2 questions. Each question carries 15 marks

17. a) Definition of an Euler path (3 marks). Explanation with example or state the conditions for a graph to have a Euler path (5 marks).
b) Example of a graph having Euler path (4 marks). Euler path. (3 marks)
[A]/[CO1]

18. x y ∇y $\nabla^2 y$ $\nabla^3 y$ $\nabla^4 y$ $\nabla^5 y$

40	184					
50	204	20				
60	226	22	2			
70	250	24	2	0		
80	276	26	2	0	0	
90	304	28	2	0	0	0

For table values (8 marks)

U= $\frac{x-xn}{h}$ = $\frac{84-90}{10}$ = 6/10 =0.6 (2 marks)

Y = $y_n + u \nabla y_{n-1} + \frac{u(u+1)}{1.2} \nabla^2 y_{n-2} +$ ……. (2 marks)

Y (84) = 321.76 ≈ 322 (substitution and final answer- (5 marks) [A]/[CO2]

19. Introducing slack variables of all constraints and rearranging equation – (2 marks)

1st iteration of simplex table construction –(8marks)

2nd iteration of simplex table creation – (3 marks) final answer: Maxi.Z =30.

Solution x_1 =0, x_2 =0, x_3 = 10 (2 marks) [A]/[CO3]

20. Verification of balanced or not – (1 mark). Convert problem into balanced (2marks) .

For finding initial feasible solution of North west corner rule method give 12marks .(type of solution is not mentioned) By using another methods. (8 marks) . [A]/[CO4]

(15X 2= 30)

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